

## Worksheet — Units, Dimensions and Vectors (Class 12 NIOS Physics)

Total marks: **30** · Questions: **10**

Instructions: Attempt all 10 questions in your notebook. Show every step of your working for the subjective questions; choose the single best option for MCQs. Answers without working will not receive full marks.

Total time: 50 minutes.

Q1. The SI unit of pressure is the: <font color='#888888'>[1 mark]</font>

- (A) newton
- (B) pascal
- (C) joule
- (D) watt

Q2. Force is a fundamental SI quantity. <font color='#888888'>[1 mark]</font>

- (A) True
- (B) False

Q3. The dimensional formula of force is  $[M L T^{\quad}]$ . <font color='#888888'>[1 mark]</font>

Q4. Which of the following has the SAME dimensional formula as work? <font color='#888888'>[2 marks]</font>

- (A) force
- (B) torque
- (C) power
- (D) pressure

Q5. The magnitude of the resultant of two perpendicular forces 8 N and 6 N is: <font color='#888888'>[2 marks]</font>

- (A) 10 N
- (B) 14 N
- (C) 2 N
- (D) 48 N

Q6. Check whether the equation  $v^2 = u^2 + 2as$  is dimensionally consistent (v, u: velocities; a: acceleration; s: displacement). Show your working. [3 marks]

Q7. Five measurements of a length give 12.4, 12.5, 12.3, 12.5, 12.4 cm. Compute the mean, the mean absolute error, and the percentage error. [3 marks]

Q8. If  $\vec{a} = 3\hat{i} + 4\hat{j}$  and  $\vec{b} = \hat{i} - 2\hat{j}$ , find (i)  $\vec{a} + \vec{b}$ , (ii)  $\vec{a} \cdot \vec{b}$ , and (iii) the magnitude  $|\vec{a} - \vec{b}|$ . [4 marks]

Q9. (a) Resolve a force of 50 N acting at  $53^\circ$  above the +x axis into x- and y-components. Take  $\cos 53^\circ \approx 0.6$  and  $\sin 53^\circ \approx 0.8$ . (b) Then find the work done on a body that moves under this force through a displacement  $\vec{d} = (4 \text{ m})\hat{i}$ . [5 marks]

Q10. (a) State the principle of dimensional homogeneity. (b) Use dimensional analysis to derive the dependence of the period T of a simple pendulum on its length L and gravitational acceleration g (up to a dimensionless constant). (c) State two limitations of dimensional analysis. [8 marks]

# Answer Key

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Q1. The SI unit of pressure is the:

**Answer: pascal**

Pa = N/m<sup>2</sup> is the SI unit of pressure.

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Q2. Force is a fundamental SI quantity.

**Answer: False**

Force is derived ( $F = ma$ ); the seven fundamentals are length, mass, time, current, temperature, amount, luminous intensity.

TRUE\_FALSE · EASY · 12-NIOS-PHY-UDV-01

Q3. The dimensional formula of force is  $[M L T^{\text{_____}}]$ .

**Answer: -2**

$F = ma \rightarrow \text{kg} \cdot (\text{m/s}^2) \rightarrow [M L T^{-2}]$ .

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Q4. Which of the following has the SAME dimensional formula as work?

**Answer: torque**

Work =  $F \cdot d \rightarrow [M L^2 T^{-2}]$ ; torque =  $r \times F \rightarrow [M L^2 T^{-2}]$ . Same dimensions despite different physical meanings.

MCQ · MEDIUM · 12-NIOS-PHY-UDV-03

Q5. The magnitude of the resultant of two perpendicular forces 8 N and 6 N is:

**Answer: 10 N**

$|R| = \sqrt{(8^2 + 6^2)} = \sqrt{(64 + 36)} = \sqrt{100} = 10 \text{ N}$ .

MCQ · MEDIUM · 12-NIOS-PHY-UDV-08

Q6. Check whether the equation  $v^2 = u^2 + 2 a s$  is dimensionally consistent (v, u: velocities; a: acceleration; s: displacement). Show your working.

**Answer:  $[v^2] = (L T^{-1})^2 = L^2 T^{-2}$ .  $[u^2] = L^2 T^{-2}$ .  $[2 a s] = (L T^{-2}) \cdot L = L^2 T^{-2}$ . All three terms have the same dimensions  $L^2 T^{-2}$ , so the equation is dimensionally consistent.**

Apply the principle of homogeneity term by term.

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Q7. Five measurements of a length give 12.4, 12.5, 12.3, 12.5, 12.4 cm. Compute the mean, the mean absolute error, and the percentage error.

**Answer:** Mean =  $(12.4 + 12.5 + 12.3 + 12.5 + 12.4)/5 = 62.1/5 = 12.42$  cm. Absolute deviations: 0.02, 0.08, 0.12, 0.08, 0.02 cm. Mean absolute error  $\Delta x = (0.02 + 0.08 + 0.12 + 0.08 + 0.02)/5 = 0.064$  cm. Percentage error =  $(0.064/12.42) \times 100 \approx 0.52\%$ .

Mean  $\rightarrow$  deviations  $\rightarrow$  mean absolute error  $\rightarrow$  divide by mean for relative, multiply by 100 for percentage.

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Q8. If  $\vec{a} = 3\hat{i} + 4\hat{j}$  and  $\vec{b} = \hat{i} - 2\hat{j}$ , find (i)  $\vec{a} + \vec{b}$ , (ii)  $\vec{a} \cdot \vec{b}$ , and (iii) the magnitude  $|\vec{a} - \vec{b}|$ .

**Answer:** (i)  $\vec{a} + \vec{b} = (3+1)\hat{i} + (4+(-2))\hat{j} = 4\hat{i} + 2\hat{j}$ . (ii)  $\vec{a} \cdot \vec{b} = (3)(1) + (4)(-2) = 3 - 8 = -5$ . (iii)  $\vec{a} - \vec{b} = (3-1)\hat{i} + (4-(-2))\hat{j} = 2\hat{i} + 6\hat{j}$ ;  $|\vec{a} - \vec{b}| = \sqrt{4 + 36} = \sqrt{40} = 2\sqrt{10} \approx 6.32$ .

Add / subtract / dot are component-wise; magnitude uses Pythagoras.

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Q9. (a) Resolve a force of 50 N acting at  $53^\circ$  above the +x axis into x- and y-components. Take  $\cos 53^\circ \approx 0.6$  and  $\sin 53^\circ \approx 0.8$ . (b) Then find the work done on a body that moves under this force through a displacement  $\vec{d} = (4 \text{ m})\hat{i}$ .

**Answer:** (a)  $F_x = F \cos \theta = 50 \times 0.6 = 30$  N.  $F_y = F \sin \theta = 50 \times 0.8 = 40$  N. So  $\vec{F} = 30\hat{i} + 40\hat{j}$  N. (b)  $W = \vec{F} \cdot \vec{d} = (30)(4) + (40)(0) = 120$  J. So 120 J of work is done.

Components first, then dot product with the displacement vector.

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Q10. (a) State the principle of dimensional homogeneity. (b) Use dimensional analysis to derive the dependence of the period T of a simple pendulum on its length L and gravitational acceleration g (up to a dimensionless constant). (c) State two limitations of dimensional analysis.

**Answer:** (a) Principle of dimensional homogeneity: in any physically valid equation every term must have the same dimensions. (b) Assume  $T = k L^a g^b$ . Match dimensions:  $T^1 = L^a (LT^{-2})^b = L^{a+b} T^{-2b}$ . T-equation:  $-2b = 1 \rightarrow b = -1/2$ . L-equation:  $a + b = 0 \rightarrow a = 1/2$ . Hence  $T \propto \sqrt{L/g}$ . The actual constant is  $2\pi$  (not fixable by dimensions alone). (c) Limitations: (i) dimensional analysis cannot determine dimensionless constants like  $2\pi$  or  $1/2$ , (ii) it cannot detect missing terms with the same dimensions as the kept terms, (iii) it cannot determine functional forms like  $\sin(x)$  or  $\exp(x)$ .

Standard pendulum-period dimensional argument; finishes with the limitations the chapter emphasises.

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